MODULE 3: ENGINEERING GRAPHICS AND CAD

(CO: 3)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with Official VIT BACEE101 Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus - All 10 Topics): (1) Engineering Drawing Introduction & Applications of CAD in Mechanical and Civil Engineering, (2) Drawing Sheet Layout, Types of Lines, Lettering Standard, Importance of Dimensioning, (3) Elements of Dimensioning, Dimensioning Systems, Rules & Practice, (4) Orthographic Projection Principles: First Angle vs. Third Angle, (5) Orthographic projection of Regular Solids in simple positions (Cube, Cylinder, Cone), (6) Orthographic projection of Prisms and Pyramids, (7) Free-hand sketching practice of simple components, (8) Isometric Projection of regular solids, (9) Perspective Projection of Cube (Visual Ray method), and (10) Drawing of a simple residential floor plan.

1 VIT FAT Exam Blueprint & Master Principles

Module 3 carries **20 to 25 marks** in BACEE101 FAT:

1. **First Angle vs Third Angle Projections (10 Marks):** Comparison table, quadrants, relative positions of object, observer, and plane, and drawing the standard projection symbols.
2. **Orthographic Views of Solids (10 Marks):** Front View (Elevation), Top View (Plan), and Side View (End view) of cubes, cylinders, cones, pyramids, and prisms with axis perpendicular or parallel to reference planes.
3. **Isometric Projections (10 Marks):** Constructing isometric scale (Isometric Length $= \sqrt{\frac{2}{3}} \times$ True Length $\approx 0.816 \times$ True Length) and drawing isometric views of cones, cylinders, and stepped blocks.

Feature	**First Angle Projection (ISO/ Indian Std)**	**Third Angle Projection (USA/AASHTO)**
Object Location	Object lies in First Quadrant (between observer and plane).	Object lies in Third Quadrant (plane between observer and object).
Plane of Projection	Opaque plane; views projected onto opposite side.	Transparent plane; views projected directly onto nearest plane.
Top View (Plan)	Drawn STRICTLY BELOW Front View.	Drawn STRICTLY ABOVE Front View.
Left Hand Side View	Drawn to the RIGHT of Front View.	Drawn to the LEFT of Front View.
Right Hand Side View	Drawn to the LEFT of Front View.	Drawn to the RIGHT of Front View.
Standard Symbol	Frustum of cone with small circle on left and trapezoid on right.	Trapezoid on left and small circle on right.

2 Drawing Standards & Dimensioning Rules

2.1 Bureau of Indian Standards (BIS / ISO) Line Types

- **Continuous Thick Line (0.5 mm):** Visible outlines and visible edges of the object.
- **Continuous Thin Line (0.2 mm):** Dimension lines, extension lines, leader lines, hatching lines for section planes.

- **Dashed Medium Line (0.3 mm):** Hidden outlines and hidden edges.
- **Chain Thin Line (0.2 mm, long dash - short dash):** Center lines, axes of symmetry, pitch circles.
- **Continuous Freehand Line (0.2 mm):** Break line for irregular boundaries.

2.2 Dimensioning Systems & Rules

1. **Aligned System:** Dimensions are placed perpendicular to the dimension line such that they can be read from the bottom or right-hand edge of the drawing sheet.
2. **Unidirectional System:** Dimensions are placed horizontally such that all values can be read exclusively from the bottom of the drawing sheet.
3. **Fundamental Dimensioning Rules:**
   - Dimension lines should never cross each other or extension lines.
   - Dimensions must be placed outside the view outlines whenever possible.
   - Never duplicate dimensions across multiple views.
   - Center lines should never be used as dimension lines.

3 Orthographic, Isometric & Perspective Projections

3.1 Orthographic Projection of Solids

- **Cube / Square Prism:** Front view is a square/rectangle; Top view is a square showing base vertices.
- **Cylinder:** Axis perpendicular to HP: Front view is a rectangle of width D and height H; Top view is a true circle of diameter D. Axis parallel to HP: Front view is a circle; Top view is a rectangle.
- **Cone:** Axis perpendicular to HP: Front view is an isosceles triangle of base D and height H; Top view is a circle with a central apex point.

3.2 Isometric Projection & Isometric Scale

An isometric projection represents a 3D object on a single 2D plane by orienting all three mutually perpendicular coordinate axes at equal angles of 120° to each other.

- The three axes are called **Isometric Axes**. Lines parallel to these axes are **Isometric Lines**.
- Foreshortening Derivation:

$$\frac{\text{Isometric Length}}{\text{True Length}} = \frac{\cos(45°)}{\cos(30°)} = \frac{\frac{1}{\sqrt{2}}}{\frac{\sqrt{3}}{2}} = \sqrt{\frac{2}{3}} \approx 0.8165$$

- In **Isometric Projection**, dimensions are drawn to the foreshortened isometric scale (81.6%). In **Isometric View (Isometric Drawing)**, true dimensions are used directly for drafting convenience.

3.3 Perspective Projection: Visual Ray Method

In perspective projection, the eye of the observer is at a finite distance (Station Point, SP) from the Picture Plane (PP). Lines drawn from the station point to different points on the object are **Visual Rays**. Where these rays pierce the picture plane determines the perspective projection. Objects farther from the observer appear proportionally smaller, closely mimicking human visual perception.